6. (E,B) to (E',B') Relativistic transformation of E and B:

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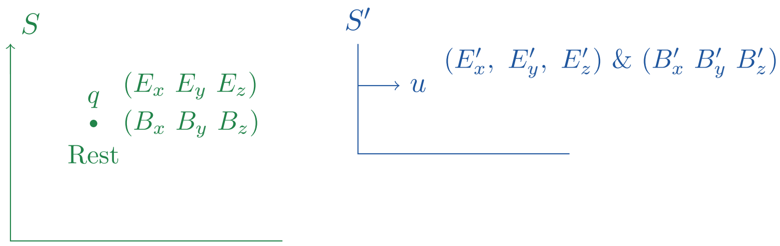
(This page includes several similer ways to develop transformation set).

let us see how Electric and magnetic fields transform under different frames of reference.



All these quantities are tensors

$$\begin{aligned} F_x &= qE_x & F'_x &= q\big(E'_x + (v' \times B')_x\big) \\ F_y &= qE_y & F'_y &= q\big(E'_y + (v' \times B')_y\big) \\ F_z &= qE_z & F'_z &= q\big(E'_z + (v' \times B')_z\big) \end{aligned}$$

$$\vec{v}\,' \times \vec{B}\,' = \begin{vmatrix} \hat{\imath}\,' & \hat{\jmath}\,' & \hat{k}' \\ -u & 0 & 0 \\ B'_x & B'_y & B'_z \end{vmatrix} \qquad (\vec{v}\,' = -\vec{u})$$

(here $\vec{v}\,'$ = velocity of particle as seen from S' frame)

$$(\vec{v}\,' \times \vec{B}\,') = \hat{\imath}\,'(0) - \hat{\jmath}\,'(-uB'_z) + \hat{k}'(-uB'_y)$$

$$(\vec{v}\,' \times \vec{B}\,')_x = 0, \qquad (\vec{v}\,' \times \vec{B}\,')_y = uB'_z, \qquad (\vec{v}\,' \times \vec{B}\,')_z = -uB'_y$$

1 (Method –1)

Now we will use transformation to see how $\vec{E}$ field & $\vec{B}$ transform

We know that,
$$F'_x = \frac{dp'_x}{dt'} = \frac{(dp_x - \beta\, dp^0)\,\gamma}{\left(dt - \dfrac{u\,dx}{c^2}\right)\gamma} = \frac{\left(F_x - \dfrac{u}{c}\right)\dfrac{1}{c}\dfrac{dE}{dt}}{\left(1 - \dfrac{uv_x}{c^2}\right)}$$

here $\dfrac{dx}{dt}$ = speed of particle from 'S' frame. $\quad (v_x = 0)$

$$\Rightarrow \qquad F'_x = \frac{\left(F_x - \dfrac{u}{c^2}\left(\vec{F}\cdot\vec{v}\right)\right)}{(1 - uv_x/c^2)} = \frac{\left(F_x - \dfrac{u}{c^2}F_x v_x\right)}{\left(1 - \dfrac{uv_x}{c^2}\right)} \qquad \begin{cases} \vec{F} = (F_x, F_y, F_z) \\ \vec{v} = (v_x, 0, 0) \\ \vec{F}\cdot\vec{v} = F_x v_x \end{cases}$$

here v = speed of charge particle in S frame $= \dfrac{dx}{dt} = v_x = 0$

$$\Rightarrow \qquad \left(F'_x \;=\; \frac{F_x - \frac{u}{c^2}(0)}{1-0} \;=\; F_x \right)$$

Similerly we have, $\quad F'_y \;=\; \dfrac{F_y\sqrt{1-u^2/c^2}}{\left(1-\dfrac{uv_x}{c^2}\right)} \;=\; \dfrac{F_y\sqrt{1-u^2/c^2}}{1-0} \;=\; \dfrac{F_y}{\gamma}$

$$F'_z \;=\; \frac{F_z\sqrt{1-u^2/c^2}}{\left(1-\dfrac{uv_x}{c^2}\right)} \;=\; \frac{F_z\sqrt{1-u^2/c^2}}{1-0} \;=\; \frac{F_z}{\gamma}$$

Now using, (1)

$$F'_x \;=\; F_x \quad \Rightarrow \quad q\big(E'_x + (v' \times B')_x\big) \;=\; qE_x$$

$$E'_x - 0 \;=\; E_x$$

$$\boxed{E'_x = E_x}$$

$$(2) \quad F'_y = \frac{F_y}{\gamma} \quad \Rightarrow \quad q\Big(E'_y + (\vec{v} \times \vec{B})_y\Big) \;=\; \frac{qE_y}{\gamma}$$

$$\Rightarrow \qquad E'_y + uB'_z \;=\; \frac{E_y}{\gamma}$$

$$\boxed{E_y \;=\; \left(E'_y + uB'_z\right)\gamma}$$

$$(3) \quad F'_z = \frac{F_z}{\gamma} \quad \Rightarrow \quad q\big(E'_z + (\vec{v} \times B')_z\big) \;=\; \frac{qE_z}{\gamma}$$

$$E'_z - uB'_y \;=\; \frac{E_z}{\gamma}$$

$$\Rightarrow \qquad \boxed{E_z \;=\; \left(E'_z - uB'_y\right)\gamma}$$

2 (2) (Method –2)

But we want to find $E'_x = f(\vec{E}, \vec{B})$, $E'_y = g(\vec{E}, \vec{B})$ & $E'_z = h(\vec{E}, \vec{B})$

To achieve it we need to just change sign of velocity of frame of reference.

$\longleftarrow$ velocity of particle in S'

$(v' = -u)$

$$F'_x \;=\; \frac{F_x - \dfrac{u}{c^2}\left(\vec{F}\cdot\vec{v}\right)}{1-\dfrac{uv_x}{c^2}} \;\Rightarrow\; F_x \;=\; \frac{F'_x + \dfrac{u}{c^2}\left(\vec{F}\,'\cdot v'\right)}{1+\dfrac{uV'_x}{c^2}} \;\Rightarrow\; F_x \;=\; \frac{F'_x - \dfrac{u^2}{c^2}F'_x}{1-\dfrac{u^2}{c^2}} \;=\; F'_x$$

$$F'_y \;=\; \frac{F_y\sqrt{1-u^2/c^2}}{1-\dfrac{uV_x}{c^2}} \;\Rightarrow\; F_y \;=\; \frac{F'_y\sqrt{1-u^2/c^2}}{1+\dfrac{uV'_x}{c^2}} \;\Rightarrow\; F_y \;=\; \frac{F'_y\sqrt{1-u^2/c^2}}{1-u^2/c^2} \;=\; \gamma F'_y$$

$$F'_z \;=\; \frac{F_z\sqrt{1-u^2/c^2}}{1-\dfrac{uV_x}{c^2}} \;\Rightarrow\; F_z \;=\; \frac{F'_z\sqrt{1-u^2/c^2}}{1+\dfrac{uV'_x}{c^2}} \;\Rightarrow\; F_z \;=\; \frac{F'_z\sqrt{1-u^2/c^2}}{1-u^2/c^2} \;=\; \gamma F'_z$$

(R.H.S should be quantities measured in S' frame)

So from our eqn set of $\vec{F}$ & $\vec{F}\,'$ we have,

$$F_x = F'_x\,, \qquad F_y = F'_y\,\gamma\,, \qquad F_z = \gamma F'_z \qquad \text{Where} \quad \gamma = \frac{1}{\sqrt{1-u^2/c^2}}$$

$$\begin{aligned} qE_x &= q\big(E'_x + (\vec{v}\,'\times B')_x\big) \\ q(E_y) &= \gamma\, q\big(E'_y + (v'\times B')_y\big) \\ q(E_z) &= \gamma\left(E'_z + (\vec{v}\,'\times \vec{B}\,')_z\right) \end{aligned}$$

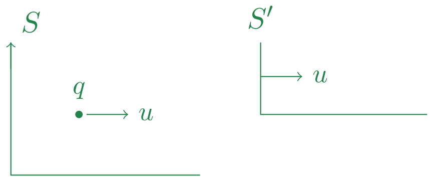
$$\boxed{E_x = E'_x + 0\,, \qquad E_y = \gamma\left(E'_y + uB'_z\right)\,, \qquad E_z = \gamma\left(E'_z - uB'_y\right)}$$

$\left(E'_x,\ E'_y,\ E'_z\right)$ we reverse the direction of moving frame $u \leftrightarrow (-u)$

$$\boxed{E'_x = E_x\,, \qquad E'_y = \gamma\left(E_y - uB_z\right)\,, \qquad E'_z = \gamma\left(E_z + uB_y\right)}$$

3 (Best way) :–

We can find these transformation from below problem



$$\begin{aligned} V = V_{\text{particle}}\Big)_S &= u\,\hat{\imath} \\ V_{S'} &= u\,\hat{\imath} \\ V_{\text{particle}}\Big)_{S'} = V' &= 0 \end{aligned}$$

$$\begin{aligned} F_x &= q\Big(E_x + (\vec{v}\times\vec{B})_x\Big) \\ F_y &= q\Big(E_y + (\vec{v}\times\vec{B})_y\Big) \\ F_z &= q\Big(E_z + (\vec{v}\times\vec{B})_z\Big) \end{aligned} \qquad \left.\begin{aligned} F'_x &= q\big(E'_x\big) \\ F'_y &= q\,E'_y \\ F'_z &= q\,E'_z \end{aligned}\right\} \quad \begin{aligned} &\text{Since } V' = 0 \\ &(V'\times B') = 0 \end{aligned}$$

using force transformation

$$F'_x = \frac{F_x - \dfrac{u}{c^2}\left(\vec{F}\cdot\vec{v}\right)}{1 - \dfrac{uV_x}{c^2}} = \frac{F_x - \dfrac{u^2}{c^2}F_x}{1-u^2/c^2} = F_x$$

$$F'_y = \frac{F_y\sqrt{1-u^2/c^2}}{1-\dfrac{uV_x}{c^2}} = \frac{F_y\sqrt{1-u^2/c^2}}{1-u^2/c^2} = \gamma F_y$$

$$F'_z = \frac{F_z\sqrt{1-u^2/c^2}}{1-\dfrac{uV_x}{c^2}} = \frac{F_z}{\sqrt{1-u^2/c^2}} = \gamma F_z$$

Above transformation implies.

$$1)\quad F'_x = F_x \quad\Rightarrow\quad qE'_x = q\Big(E_x + (\vec{v}\times\vec{B})_x\Big) = q\Big(E_x + \cancelto{0}{V_y}B_z - \cancelto{0}{V_z}B_y\Big)$$

$$\Rightarrow \qquad \boxed{E'_x = E_x}$$

$$2)\quad F'_y = \gamma F_y \quad \Rightarrow \quad qE'_y \;=\; \gamma q\Big(E_y + (\vec{v}\times\vec{B})_y\Big) \;=\; \gamma q\Big(E_y + \cancelto{0}{V_z} B_x - V_x B_z\Big)$$

$$\{\; E'_y \;=\; \gamma\,(E_y - uB_z)\;\}$$

$$3)\quad F'_z = \gamma F_z \quad \Rightarrow \quad q(E'_z) \;=\; \gamma q\Big(E_z + (\vec{v}\times\vec{B})_z\Big) \;=\; \gamma q\Big(E_z + V_x B_y - \cancelto{0}{V_y} B_x\Big)$$

$$\{\; E'_z \;=\; \gamma\,(E_z + uB_y)\;\}$$

<u>In short</u> :

$$\boxed{\left.\begin{array}{l} E'_x = E_x \\ E'_y = \gamma\,(E_y - uB_z) \\ E'_z = \gamma\,(E_z + uB_y) \end{array}\right\} \quad \text{(And)} \quad \left\{\begin{array}{l} E_x = E'_x \\ E_y = \gamma\,(E'_y + uB'_z) \\ E_z = \gamma\,(E'_z - uB'_y) \end{array}\right.}$$

$$\overrightarrow{(u \to -u)}$$

Together represents transformation rules for $\vec{E}$ from one frame to another.

If $(E_x = E_y = E_z = 0)$ we can see that $\vec{E}\,' \neq 0$

$$E'_x = 0, \qquad E'_y = -\gamma\, uB_z, \qquad E'_z = \gamma\, uB_y$$

So $\vec{E}$ is relativistic menifestation of magnetic field. and vice versa.

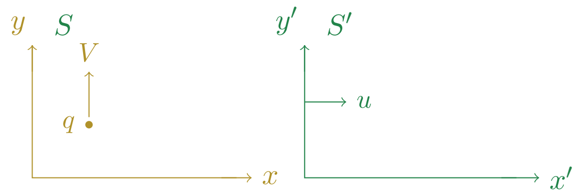
We have similar transformation of $\vec{B}$ as well.

$$\boxed{B'_x = B_x\;, \qquad B'_y = \gamma\left(B_y + \frac{u}{c^2}E_z\right)\;, \qquad B'_z = \gamma\left(B_z - \frac{u}{c^2}E_y\right)}$$

& inverse transformations

$$\boxed{(B_x = B'_x)\;, \qquad B_y = \gamma\left(B'_y - \frac{u}{c^2}E'_z\right)\;, \qquad B_z = \gamma\left(B'_z + \frac{u}{c^2}E'_y\right)}$$

4 Transformation of $\vec{B}$ to $\vec{B}\,'$:–



$$\begin{array}{ll} F_x = q(E_x + (v\times B)_x) & F'_x = q\Big(E'_x + (\vec{v}\,'\times\vec{B}\,')_x\Big) \\ F_y = q\Big(E_y + (\vec{v}\times\vec{B})_y\Big) & F'_y = q\Big(E'_y + (\vec{v}\,'\times\vec{B}\,')_y\Big) \\ F_z = q\Big(E_z + (\vec{v}\times\vec{B})_z\Big) & F'_z = q\Big(E'_z + (\vec{v}\,'\times\vec{B}\,')_z\Big) \end{array}$$

let us find $V' = ?$ $\quad V'_x \;=\; \dfrac{dx'}{dt'} \;=\; \dfrac{(dx - u\,dt)\,\gamma}{\left(dt - \dfrac{u\,dx}{c^2}\right)\gamma} \;=\; \dfrac{\left(\cancelto{0}{V_x} - u\right)}{1 - \dfrac{uV_x}{c^2}} \;=\; -u$

$$\text{similarly,}\quad V'_y = \frac{dy'}{dt'} = \frac{dy}{\left(dt - \frac{u\,dx}{c^2}\right)\gamma} = \frac{V_y\sqrt{1-u^2/c^2}}{1-\frac{uV_x}{c^2}} = V\sqrt{1-u^2/c^2}$$

$$\text{and,}\quad V'_z = \frac{dz'}{dt'} = \frac{dz}{\left(dt + \frac{u\,dx}{c^2}\right)\gamma} = \frac{\cancelto{0}{V_z}}{1-\frac{u\cancelto{0}{V_x}}{c^2}} = 0$$

$$\vec{v}\times\vec{B} = \hat{\imath}\,(V_yB_z - V_zB_y) + \hat{\jmath}\,(V_zB_x - V_xB_z) + \hat{k}\,(V_xB_y - V_yB_x)$$

$$\vec{v}\times\vec{B} = \hat{\imath}\,(V_yB_z) + \hat{\jmath}(0) + \hat{k}\,(-V_yB_x)$$

$$\vec{v}\,'\times\vec{B}\,' = \hat{\imath}\,'\big(V'_yB'_z - V'_zB'_y\big) + \hat{\jmath}\,'\big(V'_zB'_x - V'_xB'_z\big) + \hat{k}'\big(V'_xB'_y - V'_yB'_x\big)$$

$$\vec{v}\,'\times\vec{B}\,' = \hat{\imath}\,'\left(\frac{VB'_z}{\gamma}\right) + \hat{\jmath}\,'(+uB'_z) + \hat{k}'\left(-uB'_y - \frac{V}{\gamma}B'_x\right)$$

$$\begin{aligned} F_x &= q(E_x + VB_z) & F'_x &= q\big(E'_x + VB'_z/\gamma\big) \\ F_y &= q\,E_y & F'_y &= q\big(E'_y + uB'_z\big) \\ F_z &= q(E_z - VB_x) & F'_z &= q\left(E'_z - uB'_y - \frac{V}{\gamma}B'_x\right) \end{aligned}$$

use, force transformation to find relations —

$$(1)\quad F'_x = \frac{dp'_x}{dt'} = \frac{\big(dp_x - \beta\,dp^0\big)\,\gamma}{\left(dt - \frac{u\,dx}{c^2}\right)\gamma} = \frac{\left(F_x - \frac{u}{c}\frac{1}{c}\frac{dE}{dt}\right)}{\left(1 - \frac{uV_x}{c^2}\right)} = \frac{F_x - \frac{u}{c^2}\left(\vec{F}\cdot\vec{v}\right)}{\left(1 - \frac{uV_x}{c^2}\right)} = F_x - \frac{u}{c^2}F_yV_y$$

$$F'_x = F_x - \frac{u}{c^2}F_yV$$

$$\Rightarrow\qquad \left(\frac{\cancel{F'_x}}{\cancel{q}} + \frac{VB'_z}{\gamma}\right) = \cancel{F_x} + VB_z - \frac{u}{c^2}E_yV$$

$$\text{We know,}\quad \big(E'_x = E_x\big)\quad \text{So,}\quad \boxed{B'_z = \gamma\left(B_z - \frac{u\,E_y}{c^2}\right)}$$

$$(2)\quad F'_y = \frac{dp'_y}{dt'} = \frac{dp_y}{\left(dt - \frac{u\,dx}{c^2}\right)\gamma} = \frac{F_y\sqrt{1-u^2/c^2}}{1 - \frac{uV_x}{c^2}} = \frac{F_y}{\gamma}\qquad (V_x = 0)$$

$$q\big(E'_y + uB'_z\big) = \frac{qE_y}{\gamma}\qquad\Rightarrow\qquad uB'_z = \frac{E_y}{\gamma} - E'_y$$

$$B'_z = \frac{1}{u}\left(\frac{E_y}{\gamma} - \gamma\,(E_y - uB_z)\right) = \frac{1}{\gamma u}\left(E_y - \gamma^2\,(E_y - uB_z)\right)$$

$$B'_z = \frac{1}{u\gamma}\left(E_y(1-\gamma^2) + uB_z\gamma^2\right) = \frac{\gamma^2}{u\gamma}\left(E_y\left(\frac{1}{\gamma^2} - 1\right) + uB_z\right)$$

$$B'_z = \frac{\gamma}{u}\left(E_y\left(\cancel{1} - \frac{u^2}{c^2} - \cancel{1}\right) + uB_z\right)$$

$$B'_z = \frac{u}{u}\gamma\left(B_z - \frac{u\,E_y}{c^2}\right)$$

$$\boxed{B'_z = \gamma\left(B_z - \frac{u\,E_y}{c^2}\right)}$$

$$(3) \quad F'_z = \frac{dp'_z}{dt'} = \frac{dp_z}{\left(dt - \frac{u\,dx}{c^2}\right)\gamma} = \frac{F_z}{\left(1 - \frac{uV_x}{c^2}\right)\gamma} = \frac{E_z}{\gamma} \qquad \because\ (V_x = 0)$$

$$\text{So,} \qquad q\left(E'_z - uB'_y - \frac{V}{\gamma}B'_x\right) = \frac{q\,(E_z - VB_x)}{\gamma}$$

$$E'_z - uB'_y - \frac{V}{\gamma}B'_x = \frac{E_z - VB_x}{\gamma}$$

$$\gamma\,(E_z + uB_y) - uB'_y - \frac{V}{\gamma}B'_x = \frac{E_z - VB_x}{\gamma}$$

$$uB'_y + \frac{V}{\gamma}B'_x = \gamma E_z + \gamma u B_y - \frac{E_z}{\gamma} + \frac{VB_x}{\gamma}$$

$$= \frac{E_z\left(\gamma^2 - 1\right)}{\gamma} + B_y u\gamma + \frac{VB_x}{\gamma}$$

$$= \gamma E_z\left(1 - \left(1 - \frac{u^2}{c^2}\right)\right) + \cdots$$

$$\left(uB'_y + \frac{V}{\gamma}B'_x = \gamma\,E_z\,\frac{u^2}{c^2} + B_y u\gamma + \frac{VB_x}{\gamma}\right)$$